

Sehri Foods: Calm Shelves, Wild Orders

Quick Case | Supply Chain Analytics, MGT-353 | Decision-Centric Analytics

STUDENT VERSION • Reading time: approximately 12 minutes

This is a fictional case written for classroom use. Sehri Foods Limited, every person in it, and every figure are invented. The situation is modeled on distribution patterns common in Pakistani FMCG. The case intentionally does not name the analytical phenomenon you are expected to discover.

A Line That Is Either Screaming or Sleeping

On the third Monday of October, Nadia Hashmi stood on the mezzanine above Line 2 at Sehri Foods' Sheikhpura Road plant and watched nothing happen.

Three weeks earlier the same line had been running flat out, its operators working double shifts and Sunday mornings to clear a mountain of spaghetti and macaroni orders that had landed on the planners' desks in a single week. Now the extruders were silent, and the planning board showed almost nothing confirmed for the days ahead. Yet that morning's retail audit sat in Nadia's inbox, and it said what it had always said: families across Punjab were buying Sehri pasta at roughly the same weekly pace.

As Head of Supply Chain, Nadia could not shake one question:

"If households buy pasta so calmly, why does the factory behave like a patient with a fever?"

Sehri Foods

Sehri began in 1991 as a small vermicelli (seviyan) unit in rented premises on Sheikhpura Road, Lahore. Three decades later it sells 42 products, from spaghetti and macaroni to penne and instant noodles, and last year's net sales were PKR 11.4 billion. Pasta keeps for months and household consumption is normally steady, with a gentle lift before Ramadan and Eid.

The plant makes about 14,500 cartons a week on average. Regular two-shift capacity is 16,000 cartons; overtime can stretch that to 19,500. The main raw material, durum semolina, comes from flour mills in Lahore and Faisalabad in 25-tonne truckloads.

From Factory to Kitchen

Semolina Supplier → Sehri Plant → Distributor → Wholesaler/Sub-dealer → Kiriyana Shop → Household

Sehri sells through 14 independent distributors. The six Punjab distributors account for about 42% of volume and are the focus of this case. Every Monday each distributor checks stock and sends an order by WhatsApp or phone to the Sehri sales officer. Production is planned each Thursday for the following week. Delivery takes nine days on average, ranging from six to fourteen.

Distributors provide a monthly stock statement if asked; very few share daily sell-through. The most reliable market signal visible to Sehri is therefore the order placed by the distributor.

What the Numbers Showed

Nadia assembled twelve weeks of data for the six Punjab distributors: estimated household purchases from a 300-shop retail audit, distributor orders to Sehri, and semolina ordered by the plant. Exhibit 1 contains the raw weekly data. Nadia deliberately asked her analysts not to summarize the variability for her. She wanted to know what the data itself was saying.

Across twelve weeks, households bought 73,000 cartons and distributors ordered 73,100. The totals were almost identical. The weekly pattern was not. Retail demand ranged only from 5,800 to 6,400 cartons, while distributor orders ranged from 2,300 to 13,100.

At the factory, high-order weeks created overtime, expedited freight and shortages; low-order weeks created idle capacity and excess stock. Distributor stock cover ranged from 4 to 27 days within a typical month, while 6.4% of distributor order lines went unfilled.

Why Everyone Had a Reason

Every spike had an explanation, and every explanation had an owner. Sales ran a Mega Scheme—buy 20 cartons, get 1 free—for one week in every four, with additional schemes around Ramadan. A full truck of 1,200 cartons earned a 3% rebate. Sehri had raised prices three times in the previous 18 months, each time announcing the increase about eight days in advance.

Sales officers earned commission on primary sales—what Sehri invoiced to distributors—plus a quarterly bonus if the primary-sales target was met. Nothing in their pay depended on what distributors subsequently sold to shops. Across the previous four quarters, 41% of primary sales were invoiced in the final two weeks of the quarter.

Distributors also had rational reasons to protect themselves. With stock reviewed weekly and a nine-day average lead time, a stockout could last more than a week. Many padded the next order. They also bought ahead of expected price increases. Each actor was responding rationally to the rules it faced.

The Proposal

Nadia proposed a Smart Replenishment Program (SRP). Selected distributors would share daily stock and sales figures. Sehri's planning team would recommend what to ship and when, using smaller twice-weekly deliveries tied more closely to sell-through and inventory position. Four-weekly schemes would gradually give way to steadier trade margins, and sales incentives would move toward sell-through and availability.

An eight-week test at Sehri's company-owned Karachi depot produced encouraging results. Exhibit 4 shows the before-and-during comparison. Finance estimated that similar performance across the network could save PKR 150–190 million annually.

The Pushback

Tariq Mehmood, Head of Sales, objected: "My officers manage relationships. A distributor takes my call because I can bring him a scheme, a rebate, an early allocation. If Supply Chain decides the quantities, what exactly do I send my team out to do?"

Sales officers had another concern: primary-sales commission represented about a third of take-home pay for many of them. A smoother replenishment system would also cause a temporary dip while distributors worked down excess stock.

Haji Anwar, an independent Lahore distributor, was equally direct: "You want my daily stock and sales? Tomorrow you will use my own numbers to squeeze my margin. I also carry other brands. That data is not Sehri's business."

Rubina Siddiqui, the CFO, worried that the destocking transition would depress primary sales for several weeks while the board and lenders were watching quarterly results.

Thursday: The Decision

The Managing Director, Sikandar Qureshi, called Nadia in. "I believe the numbers. What I do not believe is that anything happens simply because the numbers are good. Tell me what has to change in this company, apart from the software, for this to work."

He asked Nadia to recommend one of three paths:

- A. Keep the current ordering system and invest in a better forecasting model.
- B. Roll out SRP across all 14 distributors immediately.
- C. Pilot SRP with selected independent distributors while redesigning incentives, trade terms, data-sharing rules and decision rights.

What should Nadia recommend—and what must happen first?

Discussion Questions

Q1. Where exactly does the distortion enter Sehri's supply chain? Using Exhibit 1, demonstrate analytically whether the problem originates in consumer demand or is created upstream. Then use Exhibit 2 to explain why.

Q2. Why does SRP face resistance? Whose income, authority, information or control does it threaten? Use case evidence.

Q3. Identify an equivalent distortion point in another Pakistani FMCG chain. Separate company-created distortion from channel-created distortion.

Q4. Assume the technology works. What must Nadia change in incentives, trade terms, decision rights, KPIs, data sharing and distributor relationships? What should change first, and why?

Analytics Challenge. Which demand signal should Sehri forecast? What metric(s) would you use to prove amplification, and how would you test whether schemes and price events are associated with order spikes?

Final CEO Question. Do we need a better forecasting model—or do we need to redesign the system that generates the data we are forecasting?

Terms Used in This Case

Term	Meaning
Primary sales	What Sehri invoices to distributors (sell-in).
Retail demand	What shops sell to households (sell-through); the closest available consumption signal.
Scheme	A temporary free-goods or discount offer made to distributors.
Stock cover	How many days of sales the stock on hand would last.
Coefficient of variation (CV)	Standard deviation ÷ mean; useful for comparing relative variability across series.
Bullwhip ratio	Variance of upstream orders ÷ variance of downstream demand. You may discover whether this concept applies.

Exhibit 1 — Weekly Demand and Orders, Punjab Region

Week	Event	Retail demand (cartons)	Distributor orders (cartons)	Semolina ordered (tonnes)
1		5,900	3,400	10
2	Mega Scheme	6,100	12,600	90
3		6,300	3,100	30
4		5,800	3,500	10
5		6,000	3,800	30
6	Mega Scheme	6,200	13,100	95
7		6,100	3,000	5
8	Price rise announced	5,900	9,200	70
9	New price takes effect	6,000	2,300	0
10	Mega Scheme	6,400	12,200	85
11		6,200	3,600	20
12		6,100	3,300	15
	12-week total	73,000	73,100	460

Student task: calculate the mean, sample standard deviation, variance and coefficient of variation for each series. Then calculate a bullwhip ratio using $\text{variance}(\text{Distributor Orders}) / \text{variance}(\text{Retail Demand})$.

Exhibit 2 — Commercial and Ordering Policies

Policy	How it works
Ordering	Weekly review; orders sent by WhatsApp/phone. No minimum or maximum order size.
Lead time	9 days average; range 6–14 days.
Full-truck rebate	3% off invoice for orders of 1,200 cartons or more.
Mega Scheme	Buy 20 cartons, get 1 free; one week in four plus seasonal schemes.
Price increases	Three in 18 months, averaging 7%; announced about 8 days before effective date.
Sales officer pay	Base + 0.6% commission on primary sales + quarterly primary-sales bonus.
Top-level KPIs	Sales: primary sales vs target. Supply Chain: fill rate and cost/carton. No shared inventory-stability KPI.
Quarter-end pattern	41% of each quarter's primary sales invoiced in final two weeks.
Information	Orders are reliable; monthly stock if requested; daily sell-through from very few distributors.

Exhibit 3 — Cost and Service Consequences

Annual cost (PKR million, management estimate)	Amount
Overtime and expedited freight	46
Extra warehousing and handling	28
Lost sales associated with distributor stockouts	190
Idle line time and spot-buying premium on semolina	35
Total	299
Distributor indicator	Value
Stock cover within a typical month	4–27 days (average 13)
Order lines unfilled	6.4%
Deliveries received per week	1

Analytics caution: the figures are management estimates. Do not automatically interpret every cost as causally attributable to the same mechanism.

Exhibit 4 — Karachi Depot Pilot

Measure	Before pilot	During pilot
CV of weekly orders placed on plant	58%	17%
Order lines unfilled	6.8%	2.9%
Average stock cover	15 days	9 days
Deliveries per week	1 large	2 smaller

Important: the Karachi depot is company-owned. It has no independent distributor margin, no distributor-level sales commission on its orders, and no external data-sharing concern. Decide what the pilot proves—and what it does not prove.